further experimentation and refinement. The complete results of the ternary plot analysis are presented in Figure S5 in the Supplementary Material.

The full spectrum of Peierls barrier values across the ternary compositional space offers many opportunities for further insights. For instance, it allows exploration of the barrier's variation within binary systems or specific ternary systems. The human-in-the-loop capability of the multi-agent system makes it possible to issue follow-up queries. Here, we instruct the model to plot the variation of the Peierls barrier with solute concentration for specific binary and ternary systems, including the Nb-Mo and Nb-Ta binary systems, as well as the $(NbMo)_{2x}Ta_{1-2x}$ and $(NbTa)_{2x}Mo_{1-2x}$ ternary systems, as shown in Figure 6(e).

The assistant agent identifies the relevant compositions from prior results and passes their Peierls barrier values to the coding agent for plotting, generating the plots shown in Figure 6(e). These plots reveal a non-linear variation of the Peierls barrier with solute concentration. Key insights include: (a) In the Nb-Mo system, the Peierls barrier increases with Nb concentration, peaking around 50%, before decreasing as Nb content approaches 100%. (b) In the Nb-Ta system, the Peierls barrier also increases with Nb concentration, but at a slower rate. (c) In both ternary systems, the Peierls barrier shows an increasing trend as x rises from 0 to 0.5.

The entire workflow is repeated for the solute/screw interaction energy parameter, $\Delta\tilde{E}p$, a key material property derived from potential energy changes. The problem-solving approach remains identical, but the model now uses the potential energy GNN to compute $\Delta\tilde{E}p$. The results are visualized in the ternary plot of Figure 6(d), showing non-linear variations of the interaction energy parameter with solute concentrations.

Further insights are gained by examining the variation of $\Delta\tilde{E}p$ for selected binary and ternary systems, as shown in Figure 6(g). The key findings include: (a) In the Nb-Mo system, $\Delta\tilde{E}p$ increases with Nb concentration, peaking around 30%, and then gradually decreases as Nb content continues to rise. This suggests that the interaction energy parameter is optimized at intermediate Nb concentrations. (b) In the Nb-Ta system, $\Delta\tilde{E}p$ is higher at high Nb concentrations compared to the Nb-Mo system, indicating stronger interactions at these concentrations. (c) In the $(NbTa)_{2x}Mo_{1-2x}$ system, $\Delta\tilde{E}p$ increases steadily with x and decreases as Mo content becomes minimal. (d) In the $(NbMo)_{2x}Ta_{1-2x}$ system, $\Delta\tilde{E}_p$ rises with increasing x, eventually stabilizing, suggesting a saturation point in interaction energy as Ta content diminishes.

These results highlight the complex, non-linear interactions between solute concentrations and the screw dislocation, which influence both the Peierls barrier and the solute/screw interaction energy parameter. The GNN-powered multi-agent system offers an effective framework for efficiently exploring the vast compositional space of multi-component alloys. It enables the identification of key trends and intricate behaviors, significantly enhancing the materials design process through faster and more intelligent exploration.

Yields stress in multi-component BCC alloys Solute-strengthening theories focus on the atomic-level mechanisms governing dislocation motion and their interactions with solutes, aiming to develop mechanistic models that predict temperature-dependent mechanical strength, specifically the yield stress, of materials. These models typically rely on a number of input parameters derived from first-principles calculations such as Density Functional Theory (DFT) or atomistic simulations based on empirical potentials. A prominent theory for screw dislocation strengthening in BCC alloys is the Maresca-Curtin theory [24], which requires inputs like the Peierls potential, the solute/screw interaction energy parameter, the kink formation energy, and vacancy/interstitial formation energies. However, due to the complex energy landscapes in multi-component systems, accurately computing these parameters poses a significant challenge. The newly developed GNN model addresses this by enabling the rapid and accurate computation of two critical parameters: the Peierls potential and the solute-dislocation interaction energy. This capability allows us to evaluate these parameters in mere seconds for each composition in the NbMoTa alloy space, offering a major breakthrough in efficiently predicting yield stress in multi-component alloys as a function of temperature.

The objective of this experiment is to demonstrate the potential of our multi-agent system in automating and significantly accelerating the materials design process by seamlessly integrating physics-based theoretical models with advanced DL-driven material prediction tools. In this task, the multi-agent system is assigned the calculation of yield stress for a series of alloy compositions over a broad range of temperatures, as shown in Figure 7(a). Upon receiving the task from the User, the dynamic collaboration between agents is initiated to accomplish the goal, as illustrated in Figure 7(b).

The process begins with the planning tool, which generates a detailed step-by-step plan outlining the required functions and their input parameters. The plan includes the use of GNN-powered tools to predict the Peierls barrier and the solute-dislocation interaction energy—key parameters for yield stress prediction. Additionally,